

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – III

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A) Aim:- Create methods add(), multiply(), subtract(), divide() with suitable parameters and call these methods using concept of C# delegate.

Algorithm:

STEP 1 :- START

STEP 2 :- Define a delegate Operations(int a, int b)

STEP 3 :- Create the calculator class with Add, Subtract, Multiply, and Divide methods

STEP 4 :- Create the calculator object and delegate object

STEP 5 :- Assign the method

STEP 6 :- Display the results

STEP 7 :- STOP

Code:

```
using System;
delegate void Operation(int a, int b);
class Calculator{
    public void Add(int a, int b){
        Console.WriteLine("Addition = " + (a + b));
    }
    public void Subtract(int a, int b){
        Console.WriteLine("Subtraction = " + (a - b));
    }
    public void Multiply(int a, int b){
        Console.WriteLine("Multiplication = " + (a * b));
    }
    public void Divide(int a, int b){
        if (b != 0)
            Console.WriteLine("Division = " + (a / b));
        else
            Console.WriteLine("Cannot divide by zero");
    }
}
class Program{
    static void Main(string[] args){
```

```

    Calculator c = new Calculator();
    Operation obj;
    obj = c.Add;
    obj(20, 10);

    obj = c.Subtract;
    obj(20, 10);

    obj = c.Multiply;
    obj(20, 10);

    obj = c.Divide;
    obj(20, 10);

    Console.ReadLine();
}
}

```

Output:

```

Addition = 30
Subtraction = 10
Multiplication = 200
Division = 2
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```

B) Aim:- Write a program using multicast delegate.

Algorithm:

STEP 1 :- START

STEP 2 :- Define the delegate

STEP 3 :- Create the methods of Hello() and Fine()

STEP 4 :- Create the object

STEP 5 :- Display the results

STEP 6 :- STOP

Code:

```

using System;
delegate void Message();
class Program{
    public void Hello(){
        Console.WriteLine("Hello guys!!!");
    }
    public void Fine(){
        Console.WriteLine("How are you all?? I'm fine.");
    }
    static void Main(string[] args){

```

```

    Program p = new Program();
    Message msg = p.Hello;
    msg += p.Fine;
    msg();
    Console.ReadLine();
}
}

```

Output:

```

Hello guys!!
How are you all?? I'm fine.
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```

C) Aim:- Create a class BankAccount with AccountNumber and Balance. Implement property for Balance with validation (Balance cannot be negative).

Algorithm:

STEP 1 :- START

STEP 2 :- Create a class named as BankAccount

STEP 3 :- Validate the balance so it cant be negative

STEP 4 :- Assign the account number

STEP 5 :- Display the account number and balance

STEP 6 :- STOP

Code:

```

using System;
class BankAccount{
    public int AccountNumber { get; set; }
    private double balance;
    public double Balance{
        get{
            return balance;
        }
        set{
            if (value >= 0)
                balance = value;
            else
                Console.WriteLine("Balance cannot be negative.");
        }
    }
}
class Program{
    static void Main(string[] args){
        BankAccount b = new BankAccount();
    }
}

```

```
b.AccountNumber = 1001;  
b.Balance = 5000;  
  
Console.WriteLine("Account Number: " + b.AccountNumber);  
Console.WriteLine("Balance: " + b.Balance);  
  
b.Balance = -2000;  
  
Console.WriteLine("Final Balance: " + b.Balance);  
  
Console.ReadLine();  
}  
}
```

Output:

```
Account Number: 1001  
Balance: 5000  
Balance cannot be negative.  
Final Balance: 5000  
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```